

Quantum-Enabled Signal Processing: a Method for Quantum Filter Design

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June 9, 2025

Abstract

In this note the authors propose a way to design low pass filter taps using quantum mechanics.

Consider the problem of designing a low pass filter. As shown in [1], the problem can be formulated as a minmax optimization problem. In [2], a quantum algorithm was proposed for the MaxSAT problem. Herein we propose to adapt that algorithm to the low pass filter design minmax problem. This will yield a quantum method to find the low pass filter coefficients; the minmax problem can be rewritten in QUBO form in certain cases, and from the other direction a MaxSAT problem can be rewritten in QUBO form as well.

Acknowledgments

The author thanks Prof. Kolin Paul of IIT Delhi, for suggesting image processing as an arena for quantum approaches.

References

- [1] T Parks and James McClellan. Chebyshev approximation for nonrecursive digital filters with linear phase. *IEEE Transactions on circuit theory*, 19(2):189–194, 1972.
- [2] Aman Chawla. Large language model-guided quantum maxsat algorithm design. February 2025. Zenodo Preprint. DOI:10.5281/zenodo.14876383.